STRENGTH OF MATERIALS

Thin and Thick Cylinders

Pressure vessel classification, hoop & longitudinal stress, dimensional changes, and Lame's theory for thick cylinders

Classification | Thin-Shell Stresses | Hoop & Longitudinal Stress | Dimensional Change | Lame's Theory

TOPIC 1

Classification of Cylinders

1 According to Dimensions

Vessels are classified by wall thickness t relative to shell diameter d.

Thin Shell

$$\frac{t}{d} \leq \frac{1}{15}$$

Thick Shell

$$\frac{t}{d} > \frac{1}{15} \quad (\text{or} \quad \frac{d}{t} > 15)$$

- Thin shell: wall thickness is small compared to diameter — stress is assumed uniform across the wall.
- Thick shell: wall thickness is significant — stress varies across the wall thickness (analysed using Lame's theory).

2 According to End Construction

Open End

Only circumferential (hoop) stress is induced by the fluid pressure.

Closed End

Both circumferential (hoop) stress and longitudinal stress are induced in addition to each other.

TOPIC 2

Stresses in a Thin Cylindrical Shell due to Internal Pressure

Assumptions in the Analysis

1. The effect of curvature of the cylinder wall is neglected.
2. Tensile stresses are uniformly distributed over the section of the walls.
3. The effect of the restraining action of the heads at the ends of the vessel is neglected.

Modes of Failure Under Internal Pressure

- Failure along the longitudinal section (circumferentially) — resisted by hoop stress.
- Failure across the transverse section, splitting the cylinder into two shells (longitudinally) — resisted by longitudinal stress.

Three Types of Stress Developed

Hoop / Circumferential Stress (σθ)

Acts tangentially to the circumference of the shell.

Longitudinal Stress (σL)

Acts parallel to the longitudinal axis of the shell.

Radial Stress

Too small in a thin shell — neglected in the analysis.

TOPIC 3

Circumferential or Hoop Stress

Tensile stress acting tangentially to the circumference (on the longitudinal section) of a thin cylindrical shell subjected to internal pressure.

Notation

- P = intensity of internal pressure
- d = internal diameter of shell
- l = length of the cylindrical shell
- t = thickness of the shell
- σθ = circumferential (hoop) stress

Derivation

Bursting force on a longitudinal section = pressure × projected area:

$$\text{Bursting force} = P \times d \times l$$

Resisting force in the shell walls (two sections of area t × l):

$$\text{Resisting force} = \sigma\theta \times 2t \times l$$

Equating bursting force = resisting force:

Hoop Stress

$$\sigma\theta = \frac{P\,d}{2t}$$

Required Thickness

$$t = \frac{P\,d}{2\sigma\theta}$$

TOPIC 4

Longitudinal Stress

Tensile stress acting parallel to the longitudinal axis, on the transverse (circumferential) section of a closed thin cylindrical shell subjected to internal pressure.

Derivation

Bursting force on the transverse section = pressure × cross-sectional area:

$$\text{Force} = P \times \frac{\pi}{4} d^2$$

Resisting force in the shell wall (area π d t):

$$\text{Resisting force} = \sigma_L \times \pi\, d\, t$$

With circumferential joint efficiency ηc:

$$t = \frac{P\, d}{4 \sigma_L \eta_c}$$

Longitudinal Stress

$$\sigma_L = \frac{P\, d}{4t}$$

Required Thickness

$$t = \frac{P\, d}{4 \sigma_L}$$

Key Relation

Longitudinal stress is half the hoop stress. So, design of a pressure vessel is based on the maximum stress minus the hoop stress.

$$\sigma_L = \frac{\sigma_\theta}{2}$$

Effect of Internal Pressure on Dimensions — Strains

Internal pressure changes both the diameter and length of a thin cylindrical shell. E = Young's modulus, ν = Poisson's ratio.

1 Hoop (Circumferential) Strain

$$\varepsilon_\theta = \frac{\sigma_\theta}{E} - \nu \frac{\sigma_L}{E}$$

$$\varepsilon_\theta = \frac{P d}{2 t E} \left(1 - \frac{\nu}{2}\right)$$

Hoop strain = change in diameter / original diameter:

$$\varepsilon_\theta = \frac{\delta d}{d}$$

Change in diameter:

$$\delta d = \frac{P d^2}{2 t E} \left(1 - \frac{\nu}{2}\right)$$

2 Longitudinal Strain

$$\varepsilon_L = \frac{\sigma_L}{E} - \nu \frac{\sigma_\theta}{E}$$

$$\varepsilon_L = \frac{P d}{2 t E} \left(\frac{1}{2} - \nu\right)$$

Longitudinal strain = change in length / original length:

$$\varepsilon_L = \frac{\delta L}{L}$$

Change in length:

$$\delta L = \frac{P d L}{2 t E} \left(\frac{1}{2} - \nu\right)$$

Effect of Internal Pressure on Dimensions — Volumetric Strain

The overall change in volume of the shell combines the hoop and longitudinal strains.

Volumetric strain = change in volume / original volume, expressed in terms of hoop and longitudinal strain:

$$\varepsilon V = \frac{\delta V}{V} = 2\varepsilon\theta + \varepsilon L$$

Substituting the expressions for εθ and εL:

$$\varepsilon V = \frac{P\,d}{2Et}\left(\frac{5}{4} - \nu\right)$$

Change in Volume

$$\delta V = V(2\varepsilon\theta + \varepsilon L) = \frac{P\,d}{Et}\left(\frac{5}{4} - \nu\right) V$$

Lame's Theory — Thick Cylinders

Thick cylinders have a wall thickness not smaller than 1/15 of the diameter. Radial stress varies across the thickness — maximum at the inner radius, minimum at the outer radius. Lame's theory gives the distribution of radial and hoop stress.

Assumptions

- The material is homogeneous and isotropic.
- Plane transverse sections remain plane under internal pressure.
- The material is stressed within the elastic limit (Hooke's Law).
- All fibres of the material are stressed independently, unconstrained by adjacent fibres.

$$t/d \geq \frac{1}{15}$$

a = internal radius
b = external radius
Pa, Pb = pressure at r = a, r = b

Lame's Equations

$$\sigma r = A - \frac{B}{r^2}$$

$$\sigma \theta = A + \frac{B}{r^2}$$

A and B are constants found from boundary conditions: σr = −Pa at r = a, and σr = −Pb at r = b.

General radial stress

$$\sigma r = \frac{Pa\,a^2 - Pb\,b^2}{b^2 - a^2} - \frac{a^2 b^2 (Pa - Pb)}{r^2 (b^2 - a^2)}$$

General hoop stress

$$\sigma \theta = \frac{Pa\,a^2 - Pb\,b^2}{b^2 - a^2} + \frac{a^2 b^2 (Pa - Pb)}{r^2 (b^2 - a^2)}$$

Lame's Theory — Special Cases

Longitudinal stress (closed ends, horizontal equilibrium):

$$\sigma L = \frac{Pa\,a^2 - Pb\,b^2}{b^2 - a^2}$$

Case I — Internal Pressure Only

(Pb = 0, Pa = P)

$$\sigma\theta = \frac{Pa^2}{b^2 - a^2}\left(1 + \frac{b^2}{r^2}\right)$$

$$\sigma r = \frac{Pa^2}{b^2 - a^2}\left(1 - \frac{b^2}{r^2}\right)$$

Maximum hoop stress occurs at the inner radius r = a:

$$\sigma\theta(max) = \frac{P(a^2+b^2)}{b^2 - a^2} \quad [\text{tensile}]$$

At r = a: σr = −P (compressive, = applied pressure).
At r = b: σr = 0.

Case II — External Pressure Only

(Pa = 0, Pb = P)

$$\sigma\theta = -\frac{Pb^2}{b^2 - a^2}\left(1 + \frac{a^2}{r^2}\right)$$

$$\sigma r = -\frac{Pb^2}{b^2 - a^2}\left(1 - \frac{a^2}{r^2}\right)$$

Both σr and σθ are always compressive.

$$\text{At } r=b:\ \sigma\theta = -\frac{P(b^2+a^2)}{b^2 - a^2},\ \sigma r = -P$$

At r = a: σr = 0.

$$\sigma\theta = -\frac{2Pb^2}{b^2 - a^2}$$

KEY TAKEAWAYS

Summary of Formulas

Classification

$$\frac{t}{d} \leq \frac{1}{15} \text{ (thin)}$$

Hoop stress

$$\sigma\theta = \frac{Pd}{2t}$$

Longitudinal stress

$$\sigma L = \frac{Pd}{4t} = \frac{\sigma\theta}{2}$$

Max shear stress

$$\tau max = \frac{\sigma\theta - \sigma L}{2}$$

Change in diameter

$$\delta d = \frac{Pd^2}{2tE}\left(1 - \frac{\nu}{2}\right)$$

Change in volume

$$\delta V = \frac{Pd}{Et}\left(\frac{5}{4} - \nu\right)V$$

Lame's equations

$$\sigma r = A - \frac{B}{r^2}\ ,\ \sigma\theta = A + \frac{B}{r^2}$$

Thick cyl. (internal P)

$$\sigma\theta(max) = \frac{P(a^2+b^2)}{b^2-a^2}$$

GATE QUESTION 1

A thin gas cylinder with an internal radius of 100 mm is subject to an internal pressure of 10 MPa. The maximum permissible working stress is restricted to 100 MPa. The minimum cylinder wall thickness (in mm) for safe design must be _____ kPa.

a) 10 mm
b) 12 mm
c) 8 mm
d) 15 mm

GATE QUESTION 2

A thin cylinder of 100mm internal diameter and 5mm thickness is subjected to an internal pressure of 10 MPa and a torque of 2000 Nm. Calculate the magnitudes of the principal stresses.

a) 109.75 & 40.25 MPa
b) 19.75 & 40.25 MPa
c) 101.75 & 41.25 MPa
d) 19.75 & 44.25 MPa

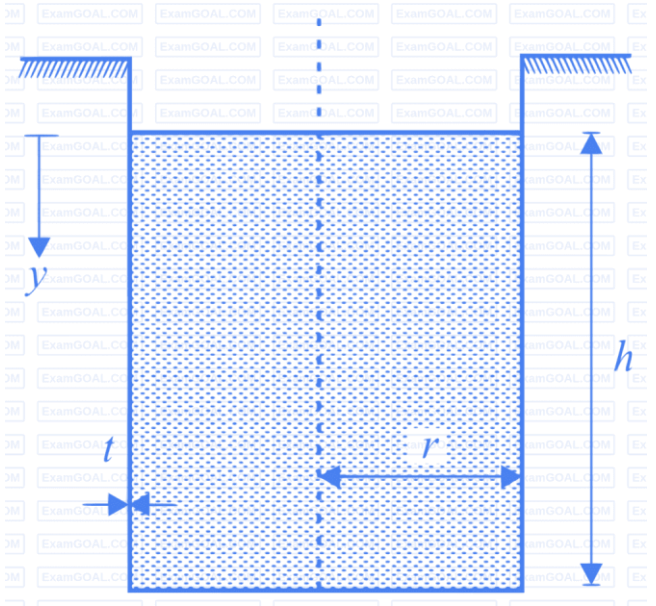
GATE QUESTION 3

A thin-walled cylindrical pressure vessel has mean wall thickness of t and nominal radius of r. The Poisson's ratio of the wall material is 1/3. When it was subjected to some internal pressure, its nominal perimeter in the cylindrical portion increased by 0.1% and the corresponding wall thickness became t. The corresponding change in the wall thickness of the cylindrical portion, i.e. 100 x (t- t)/t, is _____%.(round off to 3 decimal places).

a) -0.062 to -0.058
b) 0.062 to 0.058
c) -0.038 to -0.046
d) 0.038 to 0.046

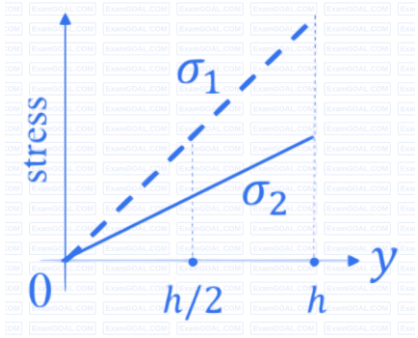
GATE QUESTION 4

The figure shows a thin-walled open-top cylindrical vessel of radius r and wall thickness t. The vessel is held along the brim and contains a constant-density liquid to height h from the base. Neglect atmospheric pressure, the weight of the vessel and bending stresses in the vessel walls.

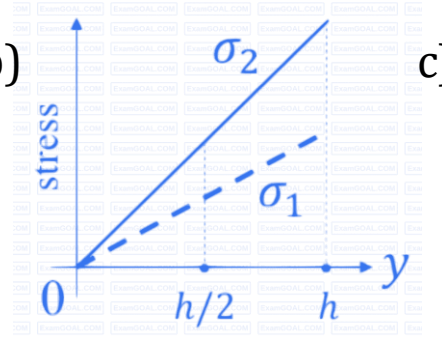


Which one of the plots depicts qualitatively CORRECT dependence of the magnitudes of axial wall stress (σ1) and circumferential wall stress (σ2) on y?

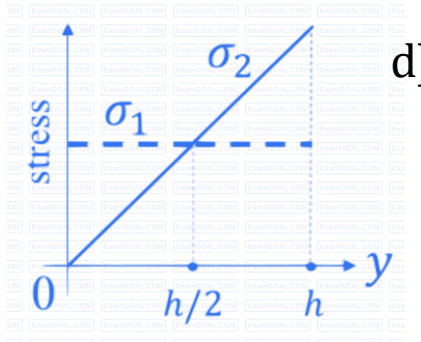
a)



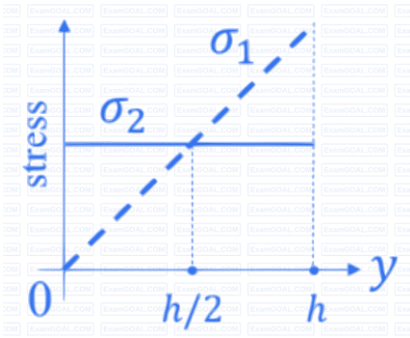
b)



c)



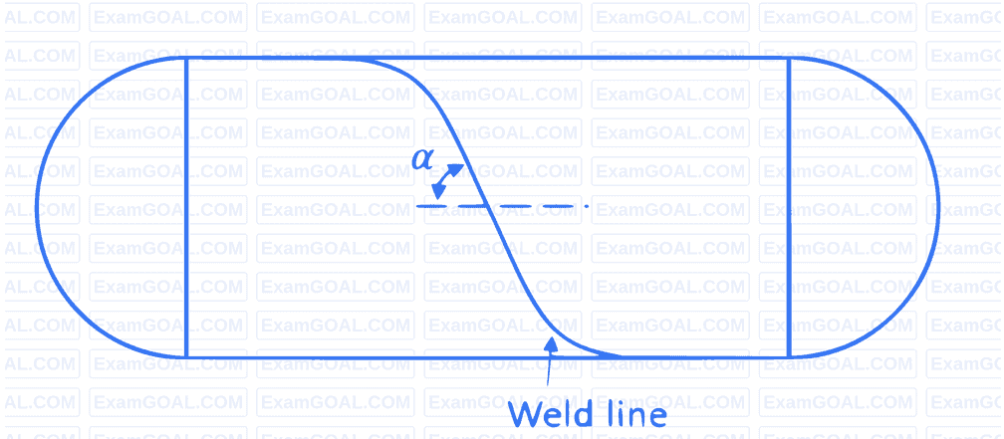
d)



GATE QUESTION 5

The figure shows a thin cylindrical pressure vessel constructed by welding plates together along a line that makes an angle a = 60° with the horizontal. The closed vessel has a wall thickness of 10 mm and diameter of 2 m. When subjected to an internal pressure of 200 kPa, the magnitude of the normal stress acting on the weld is _____MPa (rounded off to 1 decimal place).

a) 12.3 - 12.7
b) 12.0 - 12.2
c) 11.5 – 11.7
d) 13.3 – 13.5



GATE QUESTION 6

A cylindrical pressure vessel made of steel has diameter of 3 m and wall thickness of 15 mm. For steel, Young's modulus and Poisson's ratio are 210 GPa and 0.3, respectively. The cylinder is designed such that the allowable normal strain at the outer cylindrical surface is equal to 0.00034. The permissible pressure in the tank is _____kPa (rounded off to 1 decimal place).

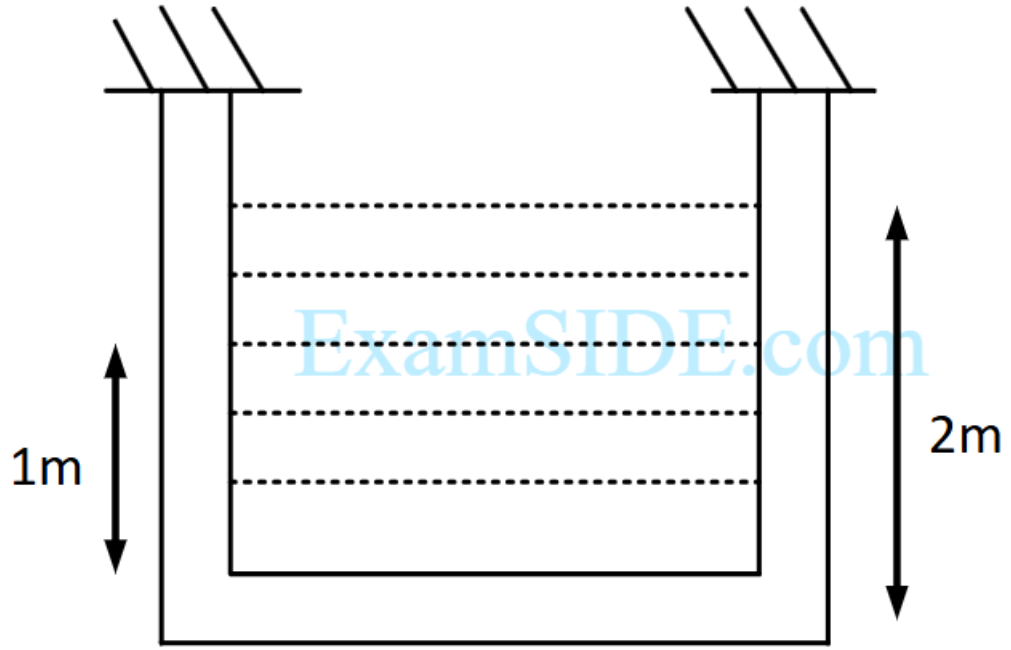
a) 840
b) 820
c) 760
d) 880

GATE QUESTION 7

A cylindrical container of radius R = 1m, wall thickness 1mm is filled with water upto a depth of 2m and suspended along with its upper rim. The density of water is 1000kg/m^3 and acceleration due to gravity is 10 m/s^2. The self weight of the cylinder is negligible. The formula for hoop stress in a thin walled cylinder can be used at all points along the height of the cylindrical container.

The axial and circumferential stress (σa, σc) experienced by the cylinder wall a mid-depth (1m as shown) are

a) (10, 5) MPa
b) (5, 10) MPa
c) (5, 5) MPa
d) (10, 10) MPa



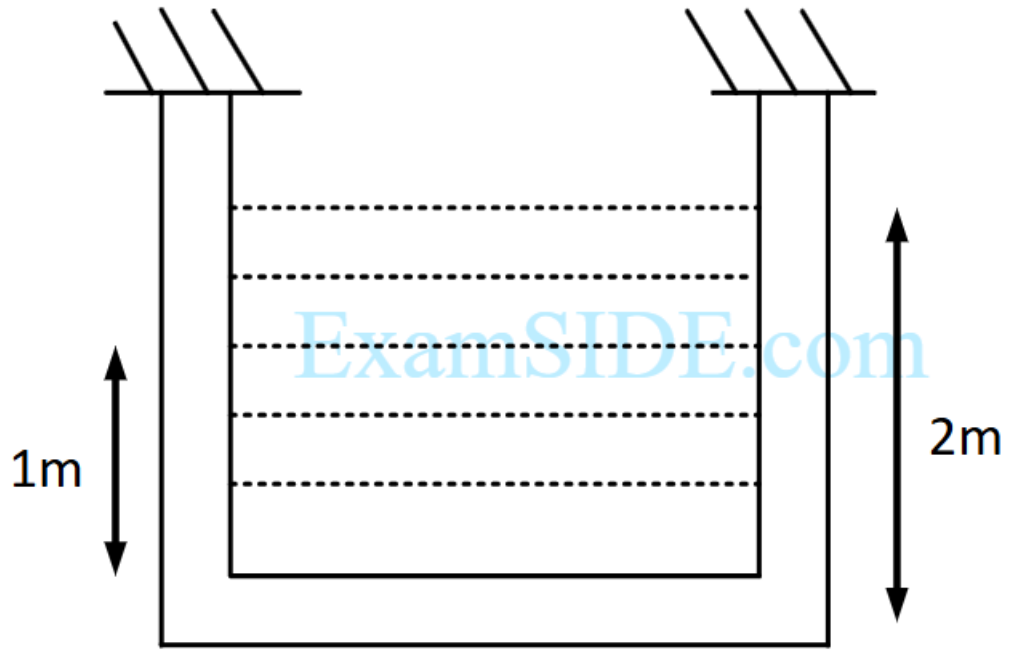
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GATE QUESTION 8

A cylindrical container of radius R = 1m, wall thickness 1mm is filled with water upto a depth of 2m and suspended along with its upper rim. The density of water is 1000kg/m^3 and acceleration due to gravity is 10 m/s^2. The self weight of the cylinder is negligible. The formula for hoop stress in a thin walled cylinder can be used at all points along the height of the cylindrical container.

If the Young's modulus and Poisson's ratio of the container material are 100GPa and 0.3, respectively. The axial strain in the cylinder wall at mid height is

a) 2 x 10^-5
b) 8 x 10^-5
c) 12 x 10^-5
d) 6 x 10^-5



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GATE QUESTION 9

A long thin walled cylindrical shell, closed at both the ends, is subjected to an internal pressure. The ratio of the hoop stress (circumferential stress) to longitudinal stress developed in the shell is

a) 0.5
b) 1
c) 2
d) 4

GATE QUESTION 10

A cylindrical tank with closed ends is filled with compressed air at a pressure of 500KPa. The inner radius of the tank is 2m and it has wall thickness of 10mm The magnitude of maximum in-plane shear stress (in MPa) is _____

a) 25
b) 45
c) 30
d) 10